

10 ft Hex Cedar Gazebo — Master Plan

Location: Saanich, BC
Style: Traditional open-rafter cedar gazebo — mortise & tenon, no metal hangers
Inspiration: Natural cedar hexagonal gazebo; 6×6 posts on concrete pads, exposed rafter system with king post hub, pronounced curved knee braces
Revision: 0 | April 2025

1. Project Overview

A full-size 10-foot hexagonal cedar gazebo with a 4:12 pitched hip roof converging at a central king post hub.

All joinery is traditional mortise and tenon secured with 3/4" white oak drawbore pegs.

No metal post hangers or joist connectors — in the tradition of timber-frame construction.

Key Specifications

Parameter	Value
Footprint	10'-0" point-to-point (5'-0" post radius)
Post spacing	5'-0" on-centre (60° intervals)
Total height at hub apex	10'-7" (10.601 ft)
Clearance at beam soffit	8'-4" (8.33 ft post height)
Beam depth	7-1/4" (4×8 nominal)
Roof pitch	4:12 (18.43°)
Hip rafter miter	28.71°
Hip rafter bevel	9.10°
Rafter overhang	12" past beam centreline
Posts	6×6 Western Red Cedar
All members	Western Red Cedar
Finish	Penofin Transparent Cedar, 2–3 coats
Foundation	12" dia. concrete caissons, 24" deep, J-bolt anchors
Location seismic / wind	Saanich BC — coastal wind ~25 psf; check E66 base spec with engineer

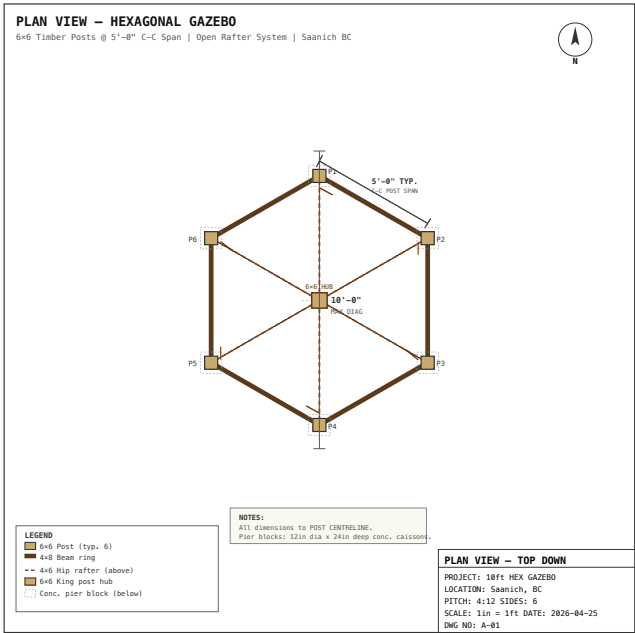
2. Architectural Drawings

All drawings produced from locked geometry (`context/staging/structural-model.json`).

Open SVG files in any modern browser for full zoom and inspection.

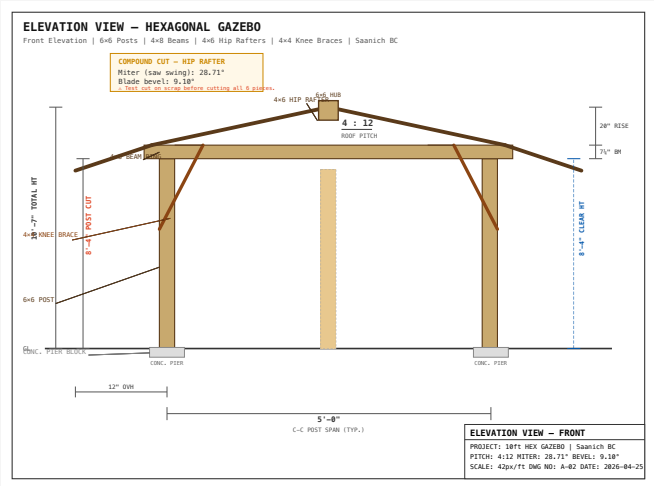
Drawing	File	Status
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Plan view (top down)

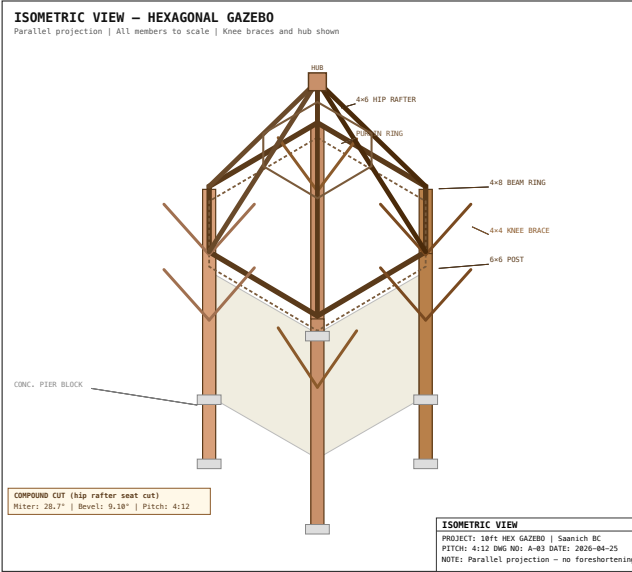
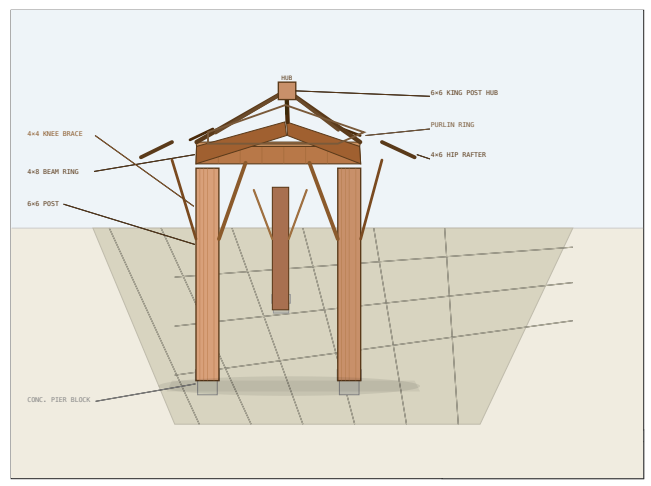


VALIDATED

Front elevation



VALIDATED

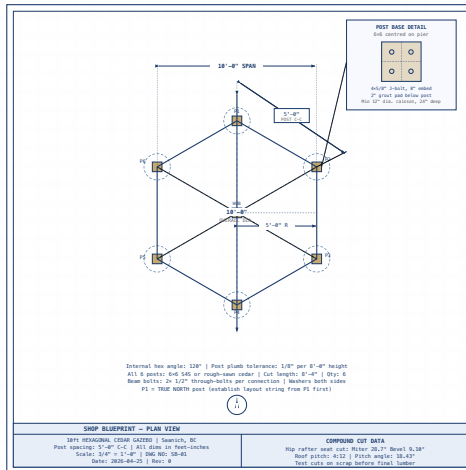
Isometric projection	ISOMETRIC VIEW – HEXAGONAL GAZEBO Parallel projection All members to scale Knee braces and hub shown  ISOMETRIC VIEW PROJECT: 10FT HEX GAZEBO Sarnich BC PITCH: 4:12 DWG NO: A-43 DATE: 2020-04-25 NOTE: Parallel projection – no foreshortening	✓ VALIDATED
Perspective view		✓ VALIDATED

3. Shop Blueprints

Dimensioned construction drawings with title blocks for shop use.

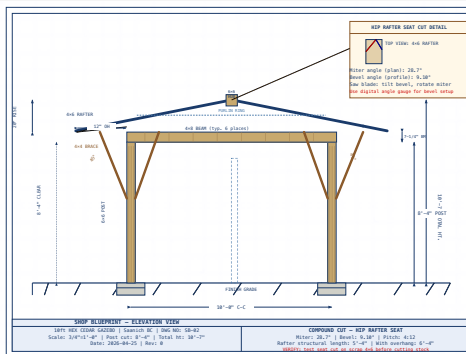
Sheet	File	Contents
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SB-01 Plan



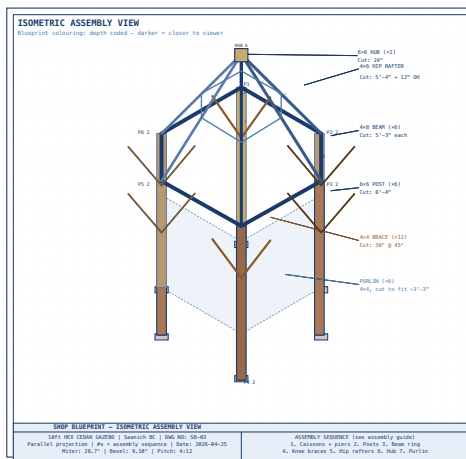
Plan with post layout, dimensions,
post-base callout

SB-02
Elevation



Elevation with compound-cut detail
balloon

SB-03
Isometric



Assembly isometric with sequence numbers 1–7

SB-04
Details

COMPONENT DETAILS - 4 DETAILS

DETAIL 1 - RSP RAFTER COMPOUND SEAT CUT

DETAIL 2 - RAFTER TENON INTO HUB MORTISE

DETAIL 3 - POST-TO-BEAM HINGED TENON

DETAIL 4 - CAISSON - POST BASE (SECTION)

COMPONENT ISOLATION - JOISTERY AND CUT DETAILS

Notes: 1. All lumber is Western Red Cedar, 2x4 minimum. 2. All joints are traditional mortise and tenon. 3. All details are shown in section. 4. All dimensions are in feet and inches. 5. All angles are in degrees. 6. All materials are to be installed as shown. 7. All materials are to be installed as shown. 8. All materials are to be installed as shown. 9. All materials are to be installed as shown. 10. All materials are to be installed as shown.

4 joinery details: seat cut, tenon, post-beam, caisson

4. Structural Members & Cut List

Full machine-readable list: [../output/shop-blueprint/SB01-cut-list.json](#)

ID	Member	Profile	Cut Length	Qty	Notes
M01	Main posts	6x6 WRC	8'-4"	6	Square ends; seal end grain
M02	Ring beam	4x8 WRC	5'-3"	6	Opposing 28.71° miters
M03	Hip rafter	4x6 WRC	5'-4" + 12" tail	6	Compound: 28.71° / 9.10°; bird's mouth at beam
M04	King post hub	6x6 WRC	2'-0"	1	6x mortises at 60° intervals
M05	Knee brace	4x4 WRC	3'-0"	12	45° compound bevel both ends
M06	Purlin ring	4x4 WRC	3'-3"	6	Half-lap to rafters mid-span

Total lumber: ~490 BF Western Red Cedar (including 10% waste)

5. Joinery Details

All joints: traditional mortise and tenon, 3/4" white oak drawbore pegs. No metal hangers or joist connectors.

Joint	Detail Sheet	Mortise Size	Peg Count
Post top → beam	SB-04 Detail 3	2.125"x4.125"×3" blind	2 pegs
Rafter → hub	SB-04 Detail 2	1.625"×3.125"×3" blind	2 pegs
Knee brace → post and beam	SB-04 Detail 3	1.5"×3"×2.5" blind	2 pegs per end

Purlin half-lap → rafter	SB-04 Detail 1 note	Half-lap depth 1.5"	1 peg per joint
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See [SB-04 component isolation sheet](#) for full dimensioned details.

6. Compound Cut Reference

Test on scrap before every production run. These angles were computed by `geometry_engine.py`.

Cut	Miter (fence)	Bevel (blade)	End
Hip rafter seat cut	28.71°	9.10°	Top/hub end
Hip rafter tail	0°	9.10°	Tail/overhang end
Ring beam ends	28.71°	0°	Both (opposing)
Knee brace ends	45°	0°	Both ends

7. Foundation

Element	Spec
Caisson diameter	12"
Caisson depth	24" below grade
Frost depth (Saanich BC)	18" nominal
Anchor	5/8"×10" J-bolt galvanised, set in wet concrete
Post base	Simpson E66 standoff (no post-in-ground)
Concrete	Quikrete 80 lb High-Strength, 2 bags per caisson

Permit note: verify setback from property lines with City of Saanich (250-475-5671) before digging.
Seismic zone 4 — confirm E66 hold-down spec with an engineer.

8. Bill of Materials Summary

Document	File
Detailed cut list (JSON)	../output/shop-blueprint/SB01-cut-list.json
Lumber purchase list	lumber-purchase-list.md
Budget estimate (CAD)	budget-estimate.md

Budget range: \$3,090 – \$4,850 CAD (DIY labour; Saanich BC April 2025 pricing)
Mid-case estimate: \$3,855 CAD

9. Assembly Sequence

Full step-by-step guide: [assembly-guide.md](#)

Phase	Task	Crew
1	Site layout + caissons	1–2
2	Post preparation + shop cuts	1
3	Erect posts + beam ring	2 min.
4	Knee braces + rafters + hub + purlins	2–3
5	Final finish + inspection	1

Estimated build time: **4–6 days** for competent DIY crew of 2 (excludes curing time).

10. Geometry Source

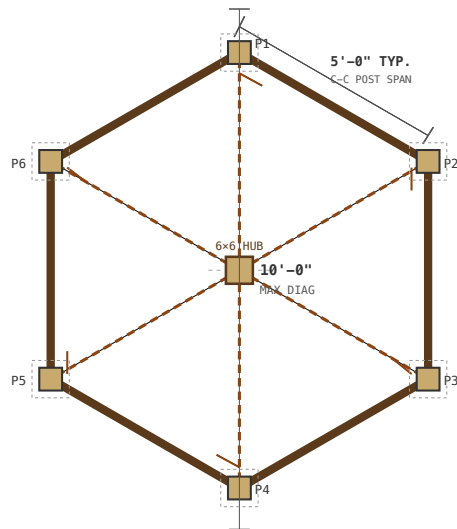
All angles, heights, and lengths derive from `plugins/garden-structure-designer/scripts/geometry_engine.py` .

Do not re-derive by hand — the engine handles the compound-angle trig correctly.

Locked model: `context/staging/structural-model.json ( _locked: true )`

```
{
  "pitch": "4:12",
  "miter_deg": 28.71,
  "bevel_deg": 9.10,
  "post_cut_ft": 8.33,
  "total_height_ft": 10.601,
  "rafter_overhang_ft": 1.0
}
```

Plan View

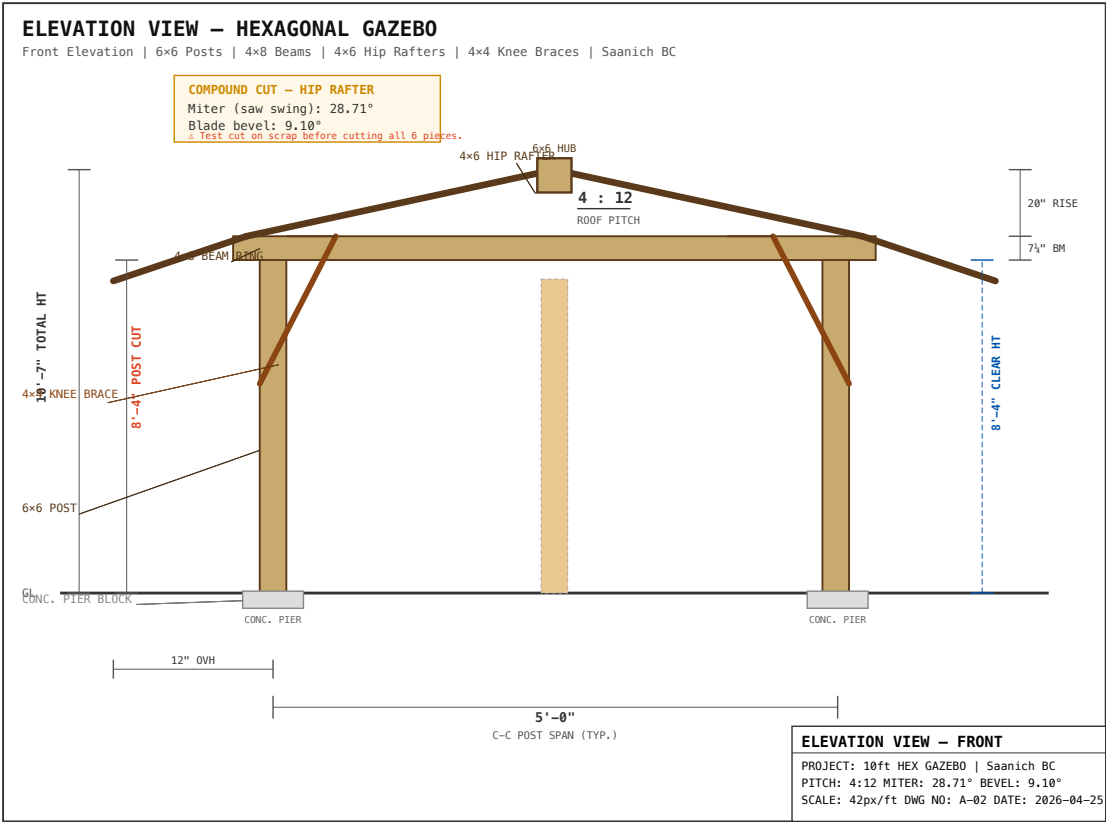


- 6x6 Post (typ. 6)
- 4x8 Beam ring
- 4x6 Hip rafter (above)
- 6x6 King post hub
- Conc. pier block (below)

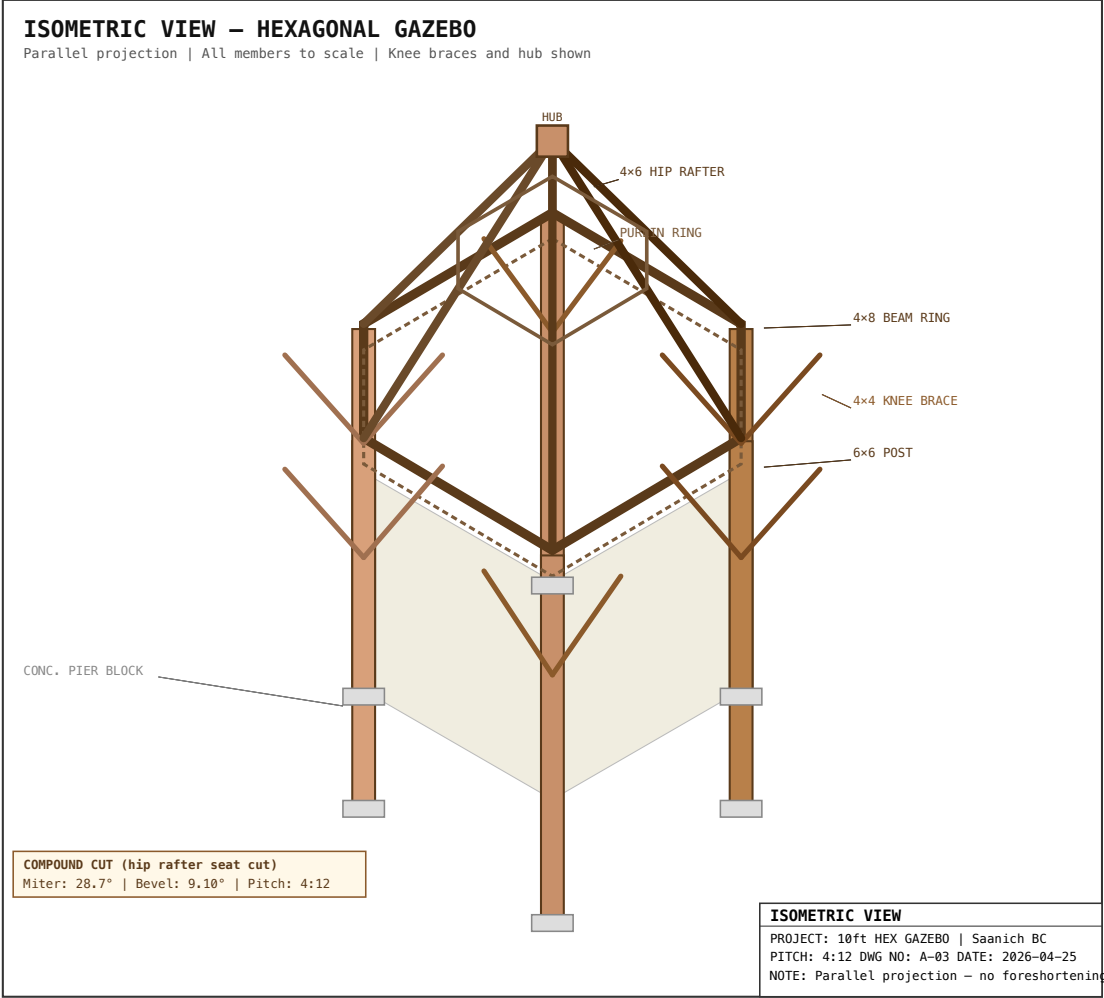
All dimensions to POST CENTRELINE.
Pier blocks: 12in dia x 24in deep conc. caissons.

PROJECT: 10ft HEX GAZEBO
LOCATION: Saanich, BC
PITCH: 4:12 SIDES: 6
SCALE: 1in = 1ft DATE: 2026-04-25
DWG NO: A-01

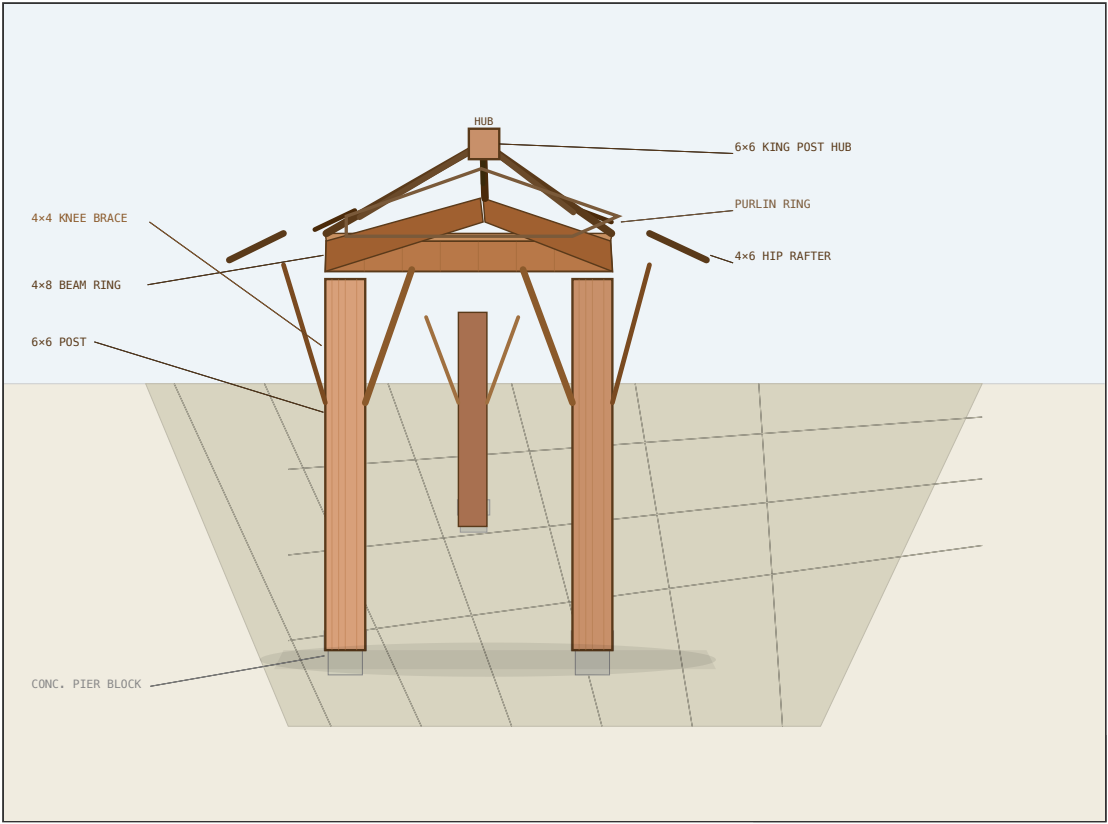
Front Elevation



Isometric Projection

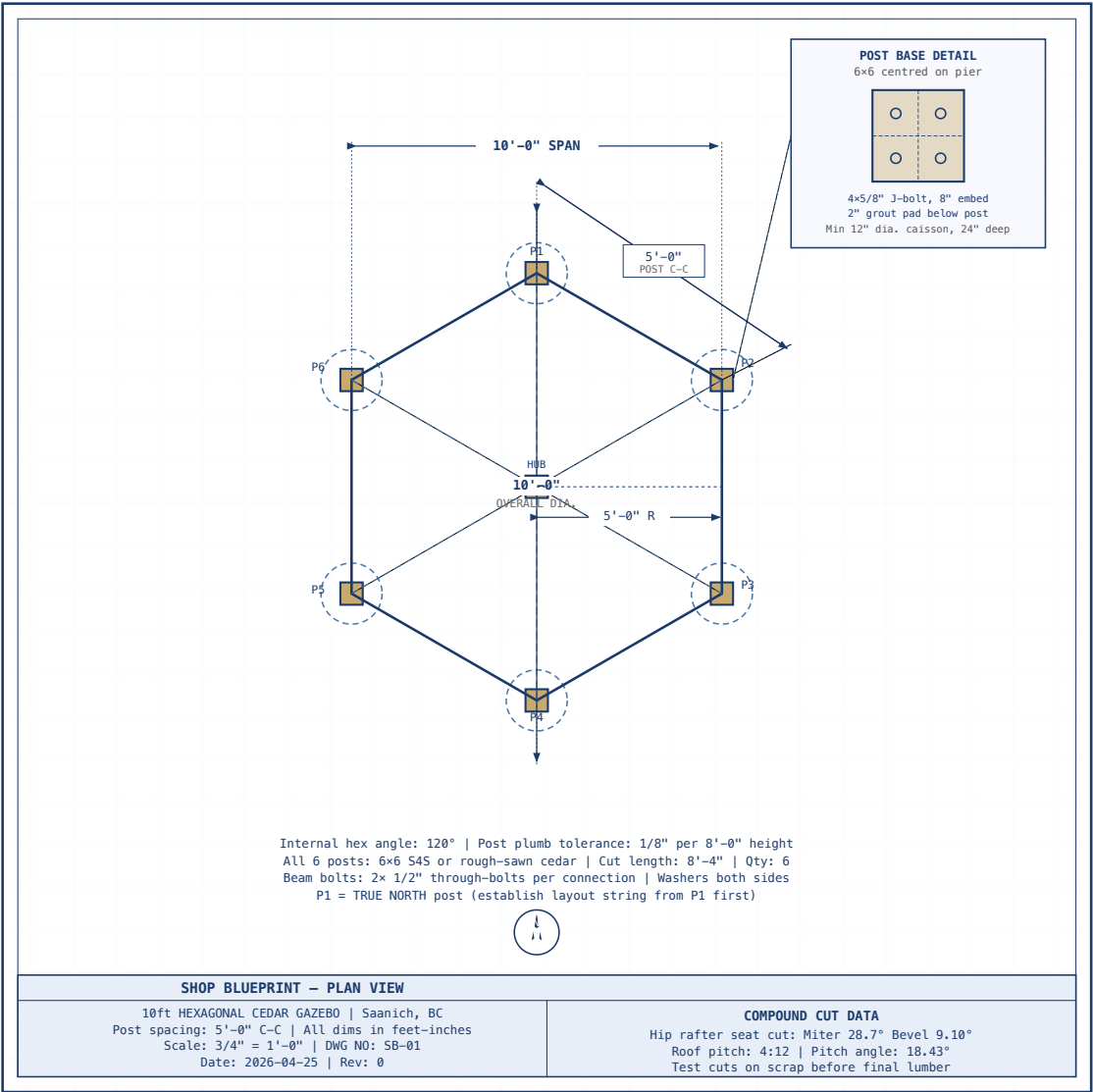


Perspective View

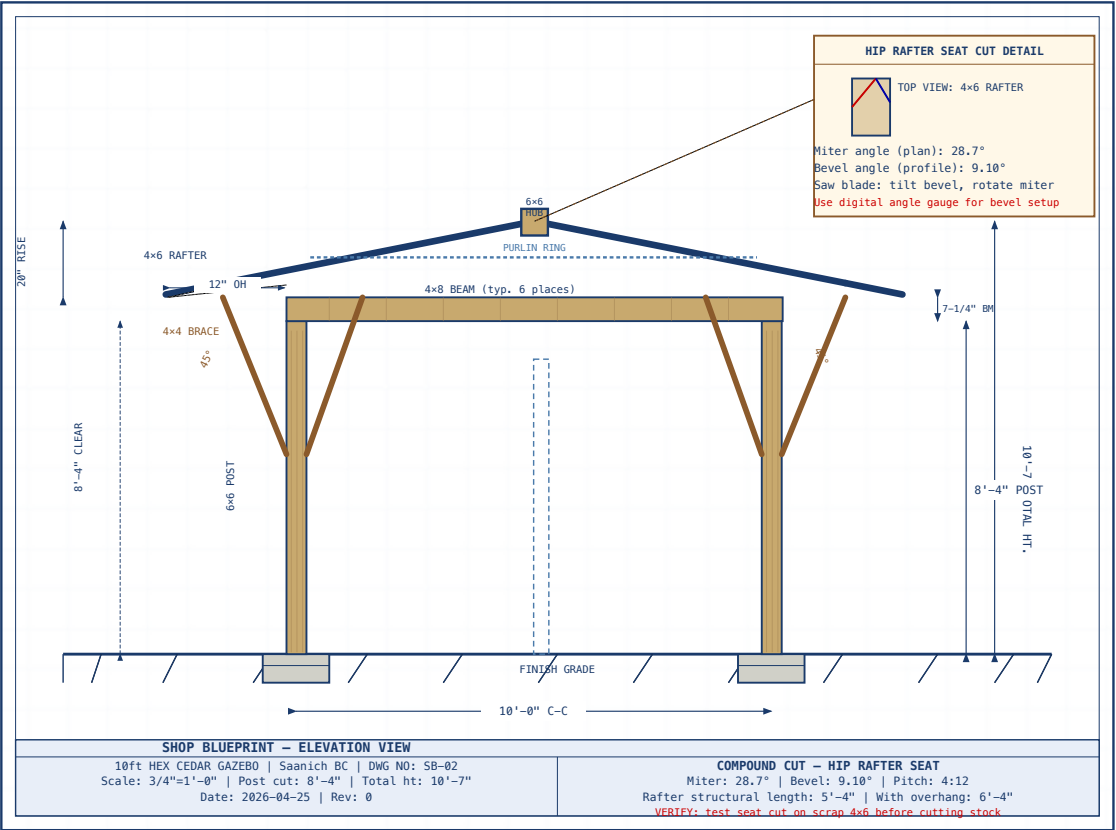


Shop Blueprints

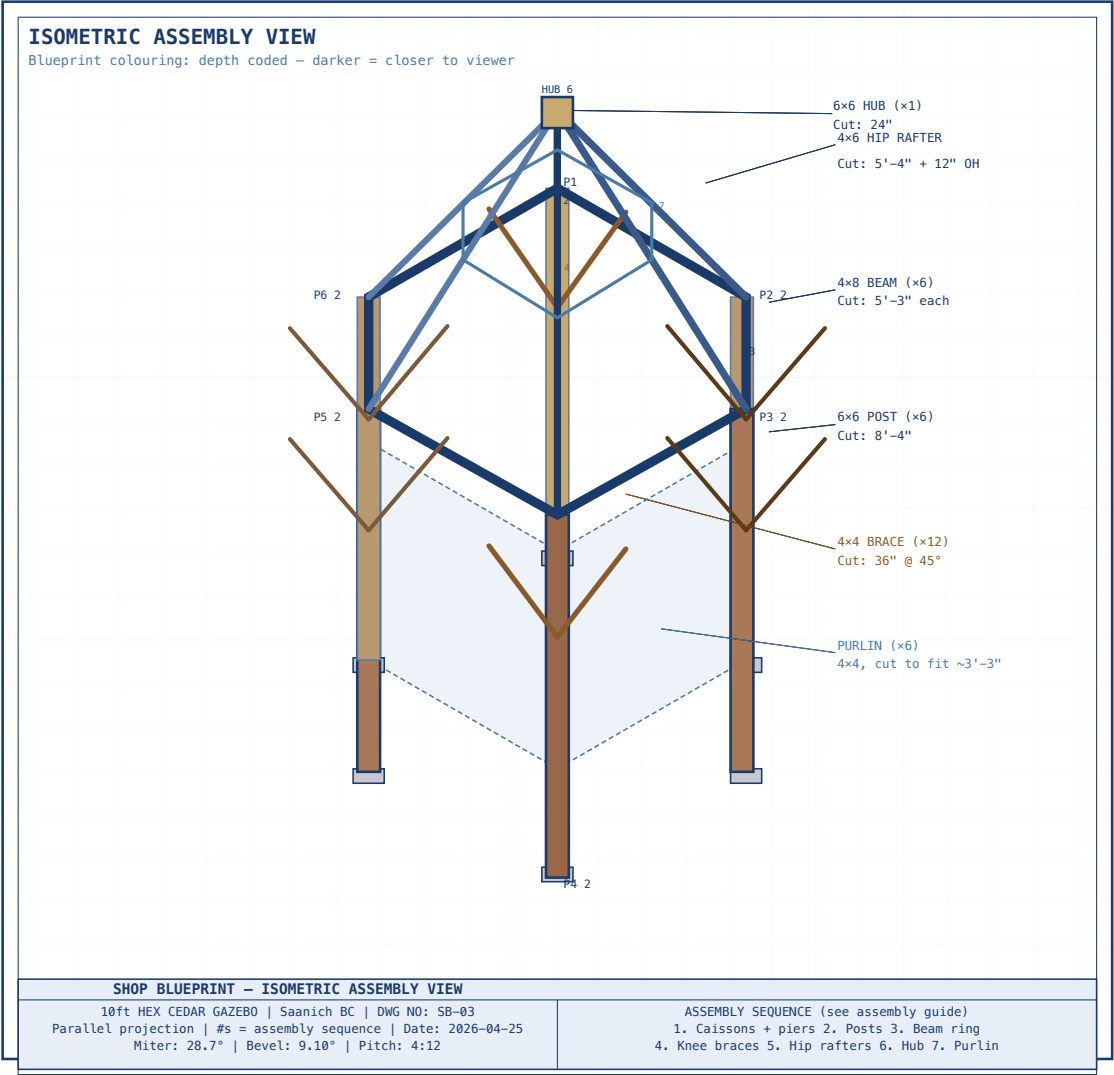
SB-01 Plan



SB-02 Elevation



SB-03 Isometric



SB-04 Details

